[Tutorial] Bitwise Operation

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- Code

```c
#include <stdio.h>
#include <stdint.h>

void dec2bin(unsigned int n);

void main() {

    uint8_t vals=0;

    // Assume 8 LEDs are connected to Digital Out pins of Port A (PA)
    // LED0 at PA[0] to LED7 at PA[7]
    uint8_t PA;

    // Initial Values of PA
    PA = 0b00001111;
    printf("\n Initial PA: ");
    dec2bin(PA);


    // Exercise_1-1: Turning ON LED4 at PA[4]
    PA |= (1<<4) ;      // turn ON LED4
    printf("\n ex1-1: ");
    dec2bin(PA);

    //Exercise_1-2:  Read the bit at PA[4], 4th from LSB
    vals =  (PA>>4)&1;  // read bit of PA[4]
    printf("\n ex1-2: %d\n", vals);
```

```c
// Exercise_2-1: Turn off LED4 at PA[4]
    PA &= ~(1<<4);      // turn off LED4
    printf("\n ex2-1: ");
    dec2bin(PA);

    //Exercise_2-2:  Read the bit at PA[4], 4th from LSB
    vals =  (PA>>4)&1;  // read bit of PA[4]
    printf("\n ex2-2: %d\n", vals);



    // Exercise_3-1: Turning ON LED5 and LED4,  at PA[5:4]
    PA |= (3<<4);   // turn ON LED5 and LED4
    printf("\n ex3-1: ");
    dec2bin(PA);

    //Exercise_3-2:  Read bits LED5 and LED4,  at PA[5:4]
    vals =  (PA>>4)&3;  // read bits at PA[5:4]
    printf("\n ex3-2: %d\n", vals);

    //Exercise_4-1:  SET  PA[0] and PA[7]
    PA |= 0b10000001;
    printf("\n ex4-1: ");
    dec2bin(PA);
```

```
    //Exercise_4-2:  CLEAR  PA[0] and PA[7]
    PA &= ~(129<<0);
    printf("\n ex4-2: ");
    dec2bin(PA);


    //Exercise_5-1:  Toggle LED7 PA[7]
    printf("\n");
    PA ^=  1<<7;
    printf("\n ex5-1: ");
    dec2bin(PA);

    printf("\n");

}

void dec2bin(unsigned int n) {
    unsigned int a = 0x80;
    for (int i = 0; i < 8; i++) {
        if ((a & n) == a)
            printf("1");
        else
            printf("0");
        a = a >> 1;
    }
}
```

- Output window

INPUT | OUTPUT | ERROR

```
Initial PA: 00001111
ex1-1: 00011111
ex1-2: 1

ex2-1: 00001111
ex2-2: 0

ex3-1: 00111111
ex3-2: 3

ex4-1: 10111111
ex4-2: 00111110

ex5-1: 10111110
```